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Structural design and aerodynamic analysis of a blended-wing-body UAV

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Abstract. Based on research findings of blended-wing-body UAVs both domestically and internationally, this paper conducts studies on parameter design, aerodynamic analysis, and structural design. A small blended-wing-body UAV is designed, providing theoretical support and technical reference for the research and development of similar UAVs.

1. Introduction

With the continuous advancement of unmanned aerial vehicle (UAV) technology, the blended-wing-body (BWB) configuration has gained increasing attention due to its superior aerodynamic performance, higher lift-to-drag ratio, and improved stealth capabilities ^[1]. Compared to conventional UAV designs, BWB UAVs offer significant advantages in terms of lift distribution, structural efficiency, and fuel economy, making them highly promising for both civilian and military applications ^[2-3].

This study builds upon domestic and international research on BWB UAVs and conducts a systematic investigation into overall design, aerodynamic analysis, and structural design. First, the initial parameters of the UAV, including maximum takeoff weight, thrust-to-weight ratio, and wing reference area, are determined, with aerodynamic layout optimization based on mission requirements ^[4]. Then, computational fluid dynamics (CFD) methods are employed to analyze the aerodynamic characteristics of the BWB UAV, evaluating its performance under different flight conditions through numerical simulations to refine the external shape ^[5-6]. Finally, in the structural design phase, suitable materials are selected according to structural design principles, and key load-bearing components undergo strength and stiffness analysis to ensure adequate structural integrity ^[7-10].

2. The Overall Design of the Blended-Wing-Body UAV

The blended-wing-body UAV is designed for aerial mapping and reconnaissance tasks. It needs to meet four main requirements: long flight range, ability to carry heavy equipment, stable flight performance, and high safety standards. The key design specifications are listed in Table 1.

Table 1. A table with design specifications.

Parameter	Requirement
Payload (W_{pl}/kg)	≥ 5
Cruising Altitude (H/m)	200
Cruising Speed $V/(m/s)$	35
Endurance (t/h)	≥ 5
Wingspan (b/m)	≤ 2



2.1. Parameter Design of the Blended-Wing Body UAV

The weight design is a fundamental aspect of the UAV's performance. The maximum takeoff weight (W_{TO}) is determined by the sum of fuel weight (W_F), empty weight (W_E), and payload weight (W_{PL}). The calculation formula is shown as Formula (1).

$$W_{TO} = W_{PL} + W_F + W_E + W_C \quad (1)$$

Aircraft weight is divided into fuel and empty weight, calculated as percentages of the total weight. These percentages are determined by using mission requirements and existing aircraft design data. Adjustments are made to balance flight time, payload capacity, and structural strength.

The thrust-to-weight ratio (how much power the engine provides compared to the aircraft's weight) affects speed, climbing ability, and agility. A higher ratio improves performance but uses more fuel. During the cruise, this ratio equals 1 divided by the lift-to-drag ratio. For takeoff, calculations show a required thrust of 65.91 N. Wing loading (takeoff weight divided by wing area) is set to 22 kg/m². This value helps achieve efficient flight by balancing weight and wing size for better aerodynamic performance.

2.2. Aerodynamic Layout Design

The UAV adopts a blended-wing-body (BWB) configuration, seamlessly integrating the wing and fuselage to reduce structural weight and minimize interference drag. This design enhances lift and achieves a high lift-to-drag ratio. However, the shorter fuselage limits the control moment arm, reducing maneuverability, especially during takeoff and landing. The design challenge lies in optimizing the aerodynamic layout, control surface configuration, and structural efficiency.

The selection of an appropriate airfoil is crucial for aerodynamic performance, influencing lift, drag, and stall characteristics. The MH108 airfoil was chosen for its excellent stall resistance and suitability for low-speed flight. It has a maximum thickness of 11.97% at 22.8% chord length and a maximum camber of 1.05% at 18.7% chord length, providing a balance between aerodynamic efficiency and structural integrity. The wing design was optimized with an aspect ratio of 2.5 to achieve a balance between aerodynamic efficiency and structural feasibility. A twist angle of 0° was adopted to simplify manufacturing while maintaining aerodynamic stability, and an incidence angle of 0° ensures smooth integration with the fuselage. Given the characteristics of the swept-wing design, a dihedral angle of 0° was selected, as it provides sufficient lateral stability without additional structural complexity.

To ensure stability and control, a twin-vertical-tail configuration was adopted. The vertical tails, canted outward, provide necessary directional stability. The NACA 0012 airfoil was selected for the vertical tails due to its strong stability and control characteristics, which are essential for maintaining flight stability. Based on the designed parameters, the UAV's final geometry was determined, with a wingspan of 1.95 m, a chord length of 0.86 m at the symmetry plane, and a reference wing area of 1.1 m². The final geometric configuration is shown in Figure 1.

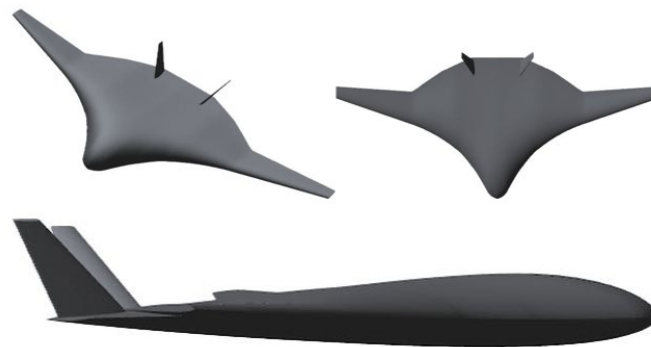


Figure 1. The geometric configuration of the blended-wing-body UAV.

2.3. Structural Design of the Blended-Wing-Body UAV

The structural design of the BWB UAV follows aerodynamic analysis, ensuring the airframe can withstand loads during flight. This design combines the wings and fuselage into a single aerodynamic shape, improving performance but adding structural challenges. The design focuses on key load-bearing components like the skin, spars, and ribs, balancing structural strength with computational efficiency. The structure must handle aerodynamic, gravitational, and propulsion forces while keeping weight low. The wing, as the main load-bearing part, generates lift and maintains stability. Two longitudinal spars, placed at 15% and 65% of the wing chord, resist bending and torsion. Ribs, spaced 300 mm apart, distribute loads and maintain the airfoil shape. The outer skin adds torsional stiffness and helps reduce weight. A beam-type wing structure was chosen for its strength, weight efficiency, and ease of manufacturing. This design also allows fuel storage within the wing, using the closed-box structure of spars, ribs, and skin to improve stiffness and save space, achieving both strength and weight reduction.

Material selection is crucial as it affects weight, strength, stiffness, and manufacturability. To achieve a high strength-to-weight ratio and withstand flight loads, carbon fiber-reinforced polymer (CFRP) was chosen as the main material. Compared to aluminum alloys, CFRP offers much higher strength and stiffness for the same weight, enabling a lighter but stronger structure. Its fiber orientation can be adjusted to match load directions, improving efficiency. The outer skin, as the main load-bearing surface, uses CFRP laminates to handle tensile and shear forces while increasing torsional stiffness. The spars and ribs are also made of CFRP to resist bending and torsion, keeping the structure lightweight yet durable.

Aluminum is easier to manufacture and costs less, but its higher density increases weight, making it less suitable for the main load-bearing structure. To balance performance and practicality, composites are used for critical components, while non-load-bearing parts may use other materials to reduce cost and ease manufacturing. Advances in automated fiber placement and resin infusion improve the precision and consistency of CFRP production. Considering these factors, the UAV structure is designed for optimal aerodynamic performance, mechanical strength, and manufacturability, ensuring it meets operational requirements while remaining lightweight. Figure 2 shows the final structural model, illustrating how the chosen materials and design enhance the UAV's performance.

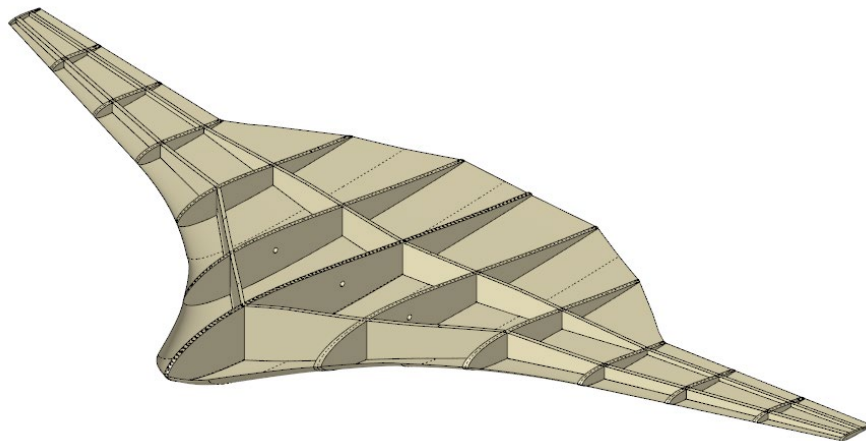


Figure 2. The structural model of the blended-wing body UAV.

3. Aerodynamic Analysis of the Blended-Wing-Body UAV

This study uses computational fluid dynamics (CFD) to analyze airflow around the blended-wing UAV. Simulations measure lift, drag, and airflow patterns at varying angles of attack. Accurate results are achieved by properly setting up the simulation area, creating detailed mesh grids, and selecting an appropriate turbulence model. The analysis provides insights into the aircraft's flight performance.

3.1. Computational Method and Boundary Conditions

This study uses ANSYS Fluent to solve the compressible Navier-Stokes formulas, as it accurately captures UAV aerodynamics under various flow conditions. Fluent is chosen for its advanced solver algorithms, reliable turbulence models, and ability to handle complex geometries. The realizable $k-\epsilon$ turbulence model is selected for its balance between accuracy and efficiency, making it suitable for external flow simulations around UAVs. The finite volume method (FVM) is used for discretization, ensuring the conservation of mass, momentum, and energy for accurate results. A second-order upwind scheme improves accuracy, especially in regions with significant aerodynamic effects. The Couple algorithm ensures stable pressure-velocity coupling and accurate convergence, particularly in complex or unsteady flow scenarios. Proper convergence criteria guarantee both numerical stability and physical accuracy. To minimize boundary effects, the computational domain is defined as a spherical region with a diameter 50 times the reference chord length, ensuring reliable results that reflect the UAV's aerodynamic performance in open-air conditions.

A polyhedral mesh is employed to enhance adaptability to complex geometries while ensuring fine resolution near the aircraft surface to capture boundary layer flow characteristics. Local mesh refinement is applied to key regions of the aircraft, and prism layers are generated along the surface to maintain y^+ values within the recommended range for the turbulence model. The computational domain is meshed with an unstructured polyhedral grid to enhance stability and reduce the number of elements. The final mesh contains approximately 4 million cells, with surface and overall mesh structures illustrated in Figure 3 and Figure 4.

The far-field region is set as a “Pressure Far-field” boundary condition, with a wind speed of 35 m/s and a temperature of 280 K. When calculating moments, the reference point is set at the aircraft's center of gravity during the cruise, located on the longitudinal symmetry plane, 0.35 m from the nose. The fluid is modeled as an ideal gas, and the wing boundaries are set as no-slip solid walls. Under the fixed wall condition, the displacement of the wall is zero, and the fluid velocity vector on the wall is also zero.

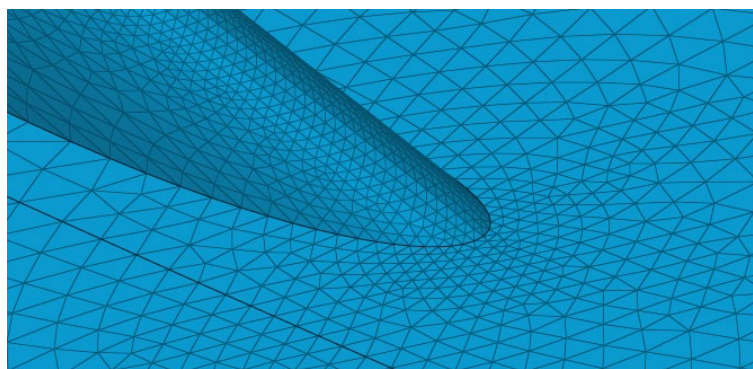


Figure 3. A close-up view of the 2D surface mesh of the blended-wing-body UAV.

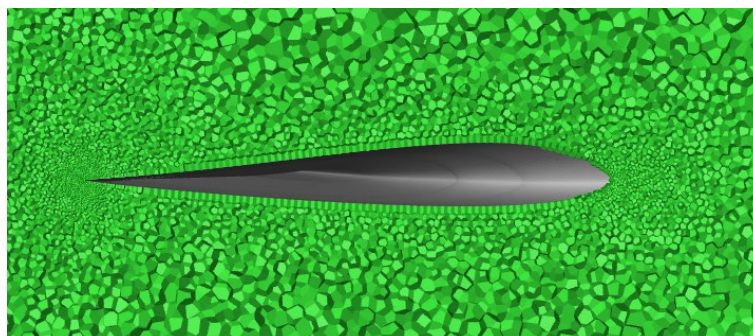


Figure 4. Local magnified view of the mesh.

3.2. Aerodynamic Performance Analysis

CFD simulations provide lift coefficient, drag coefficient, and lift-to-drag ratio at different angles of attack. Figures 5 to 8 illustrate the variation curves of various aerodynamic parameters with respect to the angle of attack (α).

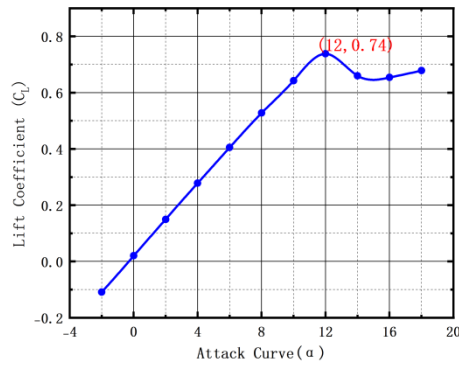


Figure 5. Lift coefficient versus angle of attack curve.

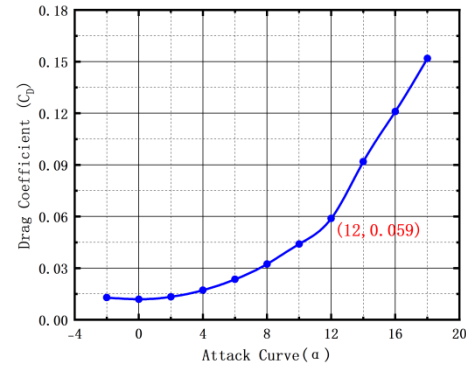


Figure 6. Drag coefficient versus angle of attack curve.

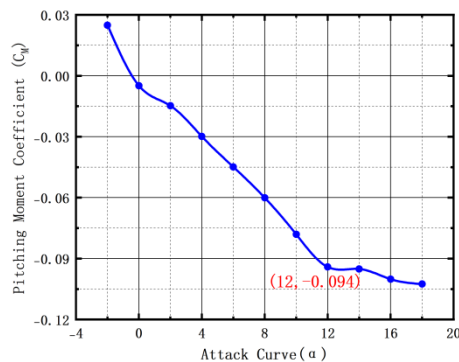


Figure 7. Pitching moment coefficient versus angle of attack curve.

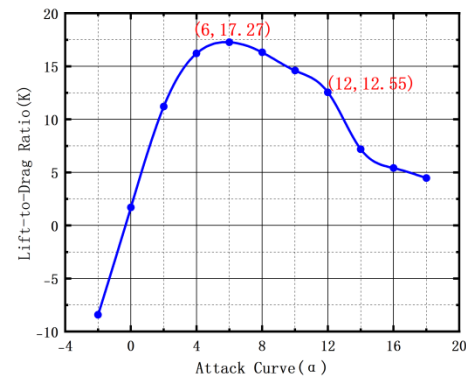


Figure 8. Lift-to-drag ratio versus angle of attack curve.

Figure 5 shows how the lift coefficient changes with the angle of attack. The lift increases until it peaks at about 0.8 around 12° , then drops and stabilizes between 0.6 and 0.65 after 14° . This means airflow starts separating beyond 12° , causing a stall, so controlling the angle is critical. In Figure 6, the drag coefficient stays steady between -2° and 4° but rises sharply after 4° , especially past 12° , confirming increased drag during stall. Figure 7 reveals the pitching moment coefficient decreases as the angle of attack grows, proving the UAV's stability. Figure 8 demonstrates the lift-to-drag ratio peaks at 17.27 when the angle is 6° , then falls as the angle increases, highlighting the blended-wing design's efficiency.

Figure 9's pressure patterns show how airflow changes with tilt angles. At lower angles, air flows smoothly over the wing, with low pressure on top creating most of the lift. At steeper angles, air starts peeling away from the wing's back edge. While lift still increases, it does so more slowly—though the design still performs well. This shows the wing shape keeps airflow attached even at extreme angles, boosting flight efficiency.

Compared to conventional UAVs with distinct fuselage and wing configurations, the blended-wing body design demonstrates superior aerodynamic efficiency. The lift coefficient (Figure 5) reaches a comparable peak but maintains smoother flow attachment beyond stall angles, reducing sudden performance drops. The drag coefficient (Figure 6) increases more gradually at higher angles, suggesting improved flow control and lower parasitic drag. Additionally, the lift-to-drag ratio (Figure 8) peaks at

17.27, which is generally higher than many traditional designs in similar flight conditions, highlighting the efficiency benefits of the blended configuration. Furthermore, the pressure distribution (Figure 9) indicates that the blended-wing shape enhances lift generation while mitigating abrupt flow separation, contributing to both aerodynamic and structural efficiency.

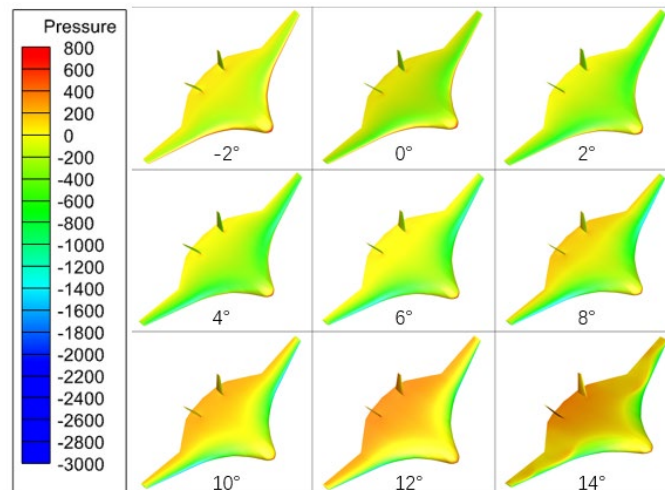


Figure 9. Pressure contour maps at different angles of attack.

4. Conclusion

This study focuses on the design and aerodynamic analysis of a small blended-wing-body (BWB) UAV, aiming to balance aerodynamic performance and structural strength. The UAV's key specifications were finalized by optimizing wing shape, proportions, and angles, strengthening its resistance to bending and twisting for stable flight and easier manufacturing. CFD tests showed efficient lift with low drag at gentle angles, but higher angles caused airflow separation, reducing performance. Airflow visualization revealed complex patterns at the wing-body junction, guiding design improvements. The UAV demonstrates better airflow handling, a stronger build, and stable flight, laying the groundwork for future advancements in aerodynamics, weight reduction, and control systems.

While CFD was used for aerodynamic analysis, the wing and fuselage shapes were not optimized in this study. Future work can refine key parameters, such as airfoil, sweep angle, and aspect ratio, through parametric modeling and optimization to enhance the lift-to-drag ratio and reduce drag. Active flow control methods, such as surface coatings and boundary layer suction, could improve performance at high angles of attack. Structurally, the current beam-based wing frame meets load requirements but offers potential for weight optimization. Topology and composite material optimization can reduce weight by refining skin, spars, and ribs. Advanced manufacturing techniques, such as 3D printing, can improve precision, minimize joints, and enhance structural integrity.

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